Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_llm_evolved_controller_diversity_failure_replay, phase8_random_portfolio.

Run Metadata

- run_name: run_20260520_233037_r
- started_at_local: 2026-05-20 23:30:37
- finished_at_local: 2026-05-21 00:01:01
- duration_hhmm: 00:30
- duration_seconds: 1824.203
- seed_offset: 17000
- replicate_label: r
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1460.617029	0.07226	0.007284	0.031222	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.222318	0.150058	275.057457	0.222318	0.093572	0.141005	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1536.636057	0.076021	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1471.485043	0.072798	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.21578	0.14352	561.677043	0.21578	0.027632	0.09695	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.214311	0.142051	612.389671	0.214311	0.045168	0.107484	0.575945	1.98	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.124043	0.051783	837.051443	0.124043	0.023034	0.060248	0.839823	2.0325	True
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	231.024057	0.213387	0.150269	0.173523	0.865434	0.56	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1460.617029 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.222318, selector regret 0.150058, runtime-adjusted gap 0.222318.
- Family transfer 0.093572 and combined transfer 0.141005.
- Runtime 275.057457 ms, runtime inflation -0.811684, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.076021.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1536.636057 ms, runtime inflation 0.052046, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072798.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1471.485043 ms, runtime inflation 0.007441, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.21578, selector regret 0.14352, runtime-adjusted gap 0.21578.
- Family transfer 0.027632 and combined transfer 0.09695.
- Runtime 561.677043 ms, runtime inflation -0.615452, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cheapest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand-2:restart2.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": -2, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-conditioned selector over frozen TSP heuristic portfolio using structure descriptors (clusteredness, bottleneck, grid-likeness, corridor/elongation, and nearest-neighbor trap). Static one-shot decision (no online switching).", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.07, "name": "tsp_heuristic_portfolio_instance_adaptive_v1", "nearest_neighbor_trap_heuristic": "nearest_neighbor_multistart", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.045168 and combined transfer 0.107484.
- Runtime 612.389671 ms, runtime inflation -0.580732, mean novelty 0.575945, and complexity 1.98.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:nearest_neighbor_multistart:stag4:cand1:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": 1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive scheduler over frozen TSP heuristic portfolio with stagnation-triggered switching and bounded schedule tuning.", "determinism": {"schedule_policy": "fully_deterministic", "seed_policy": "derive_from_instance_hash", "tie_breaker":

"lexicographic_by_portfolio_order_then_fixed_param_order"}, "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "det_tsp_hh_oracle_portfolio_v3", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "policy": {"deterministic_selection_order": ["nearest_neighbor_trap", "two_cluster_bottleneck", "corridor", "grid_like", "clustered", "default"], "online_schedule_rules": {"late_stage_refinement_preference": {"policy": "only allow stagnation_switch; avoid additional portfolio cycling; keep current heuristic otherwise.", "trigger_if": "time_budget_used >= (time_budget_trigger + 0.18)"}, "perturbation_escalation": {"escalate_to": {"else": "nearest_neighbor_trap_heuristic", "if_current_heuristic_contains": "two_opt", "then": "two_cluster_bottleneck_heuristic"}, "temperature_tuning": {"acceptance_bias": "acceptance_bias", "temperature_scale": "temperature_scale"}, "trigger_if": "failed_perturbation_count >= failed_perturbation_threshold AND time_budget_used >= time_budget_trigger"}, "stagnation_switch": {"offsets": {"candidate_limit_offset": "candidate_limit_offset if grid_likeness >= 0.25 else 0", "restart_offset": "restart_offset if time_budget_used >= 0.35 else 1"}, "switch_to": "stagnation_switch_heuristic", "trigger_if": "stagnation_length >= stagnation_threshold OR (recent_improvement_rate < low_improvement_threshold AND accepted_swaps == 0)"}}, "restart_budgeting": {"candidate_list_bias": "candidate_limit_offset if grid_likeness >= 0.25 else 0", "max_restarts": "1 + (restart_offset if stagnation_length >= 2 else 0)"}, "tests": {"clustered": {"heuristic": "clustered_heuristic", "use_if": "clustering-score >= 0.42 OR structure_class == 'clustered'"}, "corridor": {"heuristic": "corridor_heuristic", "use_if": "corridor_score >= 0.44 AND grid_likeness <= 0.18"}, "grid_like": {"heuristic": "grid_like_heuristic", "use_if": "grid_likeness >= 0.28 OR convex_hull_ratio <= 0.82"}, "nearest_neighbor_trap": {"heuristic": "nearest_neighbor_trap_heuristic", "use_if": "nearest-neighbor-trap-score >= 0.25 OR nn_distance_cv >= 0.62"}, "two_cluster_bottleneck": {"heuristic": "two_cluster_bottleneck_heuristic", "use_if": "bottleneck-score >= 0.50 OR structure_class == 'two_cluster_bottleneck'"}}, "tie_breaking": {"prefer_longer_horizon_when_time_rich": "if time_budget_used < time_budget_trigger then prefer multi-start styles: nearest_neighbor_multistart/annealed_multi_start else prefer cheaper/simpler: farthest_insertion_two_opt/candidate_pruned_two_opt", "prefer_more_specialized": true}}, "portfolio": {"heuristics": ["nearest_neighbor_trap_heuristic", "two_cluster_bottleneck_heuristic", "corridor_heuristic", "grid_like_heuristic", "clustered_heuristic", "random_like_heuristic", "default_heuristic"], "tuning_bounds": {"max_candidate_limit_multiplier": 3, "restart_multiplier_bounds": [0, 3], "temperature_multiplier_bounds": [0.6, 1.8]}}, "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "nearest_neighbor_multistart", "stagnation_threshold": 4, "temperature_scale": 1.25, "time_budget_trigger": 0.7, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.124043, selector regret 0.051783, runtime-adjusted gap 0.124043.
- Family transfer 0.023034 and combined transfer 0.060248.
- Runtime 837.051443 ms, runtime inflation -0.426919, mean novelty 0.839823, and complexity 2.0325.
- Pareto efficient: True.
- Controller signature: annealed_multi_start:cluster_first_local_search:edge_preserving_restart:farthest_insertion_two_opt:stag5:cand1:restart2.
- Controller rule summary: {"acceptance_bias": 0.009, "candidate_limit_offset": 1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive hyper-heuristic over frozen portfolio with bounded stagnation/candidate/restart/annealing adjustments; interpretable structure-driven selection with diversity-failure safeguards.", "determinism": {"seed_strategy": "deterministic_hash_from_instance_name_and_dimension", "selection_is_pure_function_of_features": true, "ties": "lexicographic_on_heuristic_roles"}, "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "det_bbmi_adaptive_portfolio_v3", "nearest_neighbor_trap_heuristic": "nearest_neighbor_multistart", "policy": {"base_selection": [{"then": "two_cluster_bottleneck_heuristic", "when": ["features.get('bottleneck_score',0.0) >= 0.6", "features.get('two_cluster_bottleneck_score',0.0) >= 0.4"]}, {"then": "clustered_heuristic", "when":

["features.get('clusteredness_score',0.0) >= 0.52", "features.get('grid_likeness',0.0) <= 0.3"]}, {"then": "grid_like_heuristic", "when": ["features.get('grid_likeness',0.0) >= 0.42", "features.get('convex_hull_ratio',0.0) >= 0.84"]}, {"then": "corridor_heuristic", "when": ["features.get('corridor_score',0.0) >= 0.45", "features.get('coordinate_spread_norm',0.0) <= 0.4"]}, {"then": "nearest_neighbor_trap_heuristic", "when": ["features.get('nearest_neighbor_trap_score',0.0) >= 0.3", "features.get('nn_distance_cv',0.0) >= 0.45"]}], "diversity_failure_mode": {"then": {"prefer_generalists_over_specialists": true, "selection_restriction": ["default_heuristic", "grid_like_heuristic", "two_cluster_bottleneck_heuristic"]}, "when": ["context.get('replay_mode','') == 'diversity_failure'", "context.get('failure_archive_size',0) >= 6"]}, "fallback": "default_heuristic", "tie_break": {"deterministic_order": ["default_heuristic", "grid_like_heuristic", "two_cluster_bottleneck_heuristic", "clustered_heuristic", "corridor_heuristic", "nearest_neighbor_trap_heuristic"]}}, "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "schedule": {"stagnation_rule": {"edge_skew_sensitive_bias": {"if_conditions": ["online.get('edge_length_skew_in_current_tour',0.0) >= 0.65"], "then": {"avoid_random_like_thrashing": true, "prefer_two_opt_stability": true}}, "perturbation_escalation": {"if_failed_perturbations": ["online.get('failed_perturbation_count',0) >= failed_perturbation_threshold"], "then": {"random_like_heuristic": "random_like_heuristic", "use_random_like_phase": true}}, "switch_to": "stagnation_switch_heuristic", "trigger_condition": ["online.get('stagnation_length',0) >= stagnation_threshold", "online.get('recent_improvement_rate',0.0) <= low_improvement_threshold"]}, "time_rule": {"then": {"annealing_adjustment": {"acceptance_bias": "acceptance_bias", "temperature_scale": "temperature_scale"}, "candidate_limit_offset": "candidate_limit_offset", "final_refinement_bias": true, "restart_offset": "restart_offset"}, "trigger_condition": ["online.get('time_budget_used',0.0) >= time_budget_trigger"]}}, "stagnation_switch_heuristic": "farthest_insertion_two_opt", "stagnation_threshold": 5, "temperature_scale": 1.25, "time_budget_trigger": 0.68, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 231.024057 ms, runtime inflation -0.841831, mean novelty 0.865434, and complexity 0.56.
- Pareto efficient: True.

Judge Appendix

Summary of TSP Adaptive Heuristic Portfolio Suite Results

Condition	Held-out TSPLIB Gap	Selector Regret vs Oracle	Runtime-adjusted Gap	Runtime Inflation	Pareto Efficient	Transfer Gap	Interpretation Highlights
phase8_best_single_fixed_heuristic (best_tsplib)	**0.07226**	0.0	0.07226	0.0	Yes	0.031222	Best *static* fixed heuristic baseline (farthest_insertion_two_opt). No runtime overhead. Interpretable baseline for adaptive selectors.
phase8_oracle_selector (best_transfer)	0.07226	0.0 (definition)	0.07602	+5.2%	No	0.026678	Non-deployable upper bound, sets target for selectors. Slight runtime inflation noted.

Condition	Held-out TSPLIB Gap	Selector Regret vs Oracle	Runtime-adjusted Gap	Runtime Inflation	Pareto Efficient	Transfer Gap	Interpretation Highlights
phase8_supervised_ml_selector	0.07226	0.0	0.07280	+0.74 %	No	0.031222	Matches best fixed heuristic in held-out gap/runtime-adjusted gap with minimal runtime inflation. Interpretable ML-based control without runtime bloat.
phase8_llm_static_selector	0.21578	0.144	0.21578	-61.5%	No	0.09695	Instance-conditioned static selector with substantial runtime savings but large gap increase, inferior to fixed heuristic baseline on primary endpoint.
phase8_llm_evolved_controller_diversity_failure_replay	0.12404	0.052	0.12404	-42.7%	Yes	0.060248	Adaptive controller with interpretable, structure-driven rules. Runtime reduced versus fixed heuristics but TSPLIB gap notably worse than fixed baseline. Some selector regret present but significantly lower than naive selectors. Reasonable transfer capability.
phase8_random_portfolio	0.22232	0.150	0.22232	-81.2%	Yes	0.141005	Naive baseline with poor gap but lowest runtime and Pareto efficient. Highest selector regret.
phase8_llm_evolved_adaptive_controller	0.21431	0.142	0.21431	-58.1%	No	0.107484	Complex adaptive controller with runtime savings but worse primary endpoint (TSPLIB gap) than fixed heuristic. Selector regret comparable to diversity-failure replay condition.
phase8_full_solver_evolution	0.21339	0.141	0.21339	-84.2%	Yes	0.173523	Evolved full solver with highest novelty but poor gap performance; Pareto efficient due to strong runtime reduction. Selector regret high.

Conservative Interpretation & Prioritized Conclusions

1. **Primary Endpoint (Held-out TSPLIB Gap):**

- The best fixed heuristic (farthest_insertion_two_opt) achieves the lowest held-out TSPLIB gap (0.07226), matched by the oracle and supervised selectors.
- Adaptive controllers reduce runtime significantly but at the cost of doubling or tripling TSPLIB gap (~0.12-0.21).
- Naive baselines (random portfolio, full solver evolution) have large gaps, not competitive for primary endpoint.

2. **Selector Regret vs Oracle:**

- Both the supervised selector and best fixed heuristic have zero selector regret (oracle-level performance).
- Adaptive controllers show modest selector regret (~0.05-0.14), indicating some suboptimal selections relative to the oracle.
- Random portfolio has highest regret (~0.15).

3. **Runtime-adjusted Gap & Runtime Inflation:**

- Best fixed heuristic runs in ~1460 ms with no inflation, achieving optimal gaps.
- Oracle selector modestly inflates runtime (+5%), supervised selector minimal (~+0.7%).
- Static LLM selector and adaptive controllers reduce runtime by ~40-60% but lose significant gap performance (trade-off).
- Random portfolio and full solver reduce runtime by ~80% but with severely worse gap.

4. **Pareto Efficiency:**

- The Pareto front includes best fixed heuristic, random portfolio, diversity-failure replay adaptive controller, and full solver.
- Oracle and supervised selectors do not belong to Pareto front likely due to runtime inflation or complexity.

5. **Cross-family Transfer:**

- Transfer gaps generally align with main gap results; best fixed heuristic and supervised selector transfer well (~0.03 gap).
- Adaptive controllers show moderate transfer gaps (~0.06-0.1), indicating some robustness but not yet matching oracle-level transfer.
- Random portfolio and full solver have poor transfer.

6. **Interpretable Instance-adaptive Control:**

- The **phase8_llm_evolved_controller_diversity_failure_replay** condition provides a good balance:
- Interpretable, deterministic structure-driven adaptive control.
- Significant runtime savings (~43%) compared to fixed heuristic.
- Moderate increase in held-out gap (0.12 vs 0.07).
- Low selector regret relative to oracle (0.05).
- Pareto efficient, indicating a meaningful trade-off.
- The supervised selector matches oracle-level gap with minimal overhead but is not Pareto efficient and less runtime-efficient.
- The purely static LLM selector has too large gap degradation despite runtime gains.

Final Recommendations

- For practical deployment prioritizing best solution quality, **the best single fixed heuristic (farthest_insertion_two_opt) or supervised ML selector** are preferred due to oracle-level performance and minimal runtime overhead.

- If runtime is constrained and interpretable adaptive control is desired, the **evolved adaptive controller with diversity-failure replay** represents a reasonable compromise, offering runtime savings with tolerable TSPLIB gap increase and low selector regret.
- Avoid overclaiming new algorithm discoveries; improvements are due to **instance-adaptive hyper-heuristic control over known heuristics**.
- The oracle selector remains a useful theoretical upper bound but is not deployable.
- Further tuning may focus on closing the gap-runtime trade-off, improving transfer robustness, and refining interpretable adaptation without runtime inflation.